

# Dhruv Malhotra

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## SUMMARY

Software Engineer in Test with 6+ years building enterprise-scale test automation, API validation frameworks, and performance benchmarking tools across cloud-native platforms (AWS, Docker, Kubernetes). Proven expertise in Python backend engineering, high-coverage testing, and CI/CD pipelines for SaaS products serving millions of users. Security clearance eligible veteran with a track record of bridging QA and backend engineering to drive system reliability and business outcomes.

## SKILLS

**Programming/ML:** Python, Java, Django, PyTorch, Pandas, TensorFlow, Flask, React.js, C/C++, Swift

**Cloud/Tools:** AWS (EC2, Lambda, S3, CloudWatch), Docker, Kubernetes, Jenkins, GitHub Actions, Git, Postman, MySQL, PostgreSQL, Redis, Browserstack

**QA/Testing:** pytest, pytest-django, Selenium, Playwright, Postman, JUnit, k6, BDD, TDD, JIRA, REST API Testing, Performance Testing

**Monitoring:** Grafana, DataDog, Prometheus

## WORK EXPERIENCE

### Software Engineer in Test, Eagle Eye Networks

Austin TX | October 2024 – Present

- Developed AI-assisted QA methodology compressing manual API testing from days to ~15 minutes via parallel automation and automated data analysis across enterprise API endpoints
- Designed and executed end-to-end test suites for GRACE error handling/retry system, validating cooloff behavior, retry policies, and webhook concurrency on EEN v3 API
- Analyzed 3,100+ alerts and 10,100+ notifications to identify architectural patterns in GRACE cooloff behavior, discovering a shared rule-level mechanism with fixed 10-minute intervals that informed company design decisions
- Built a fault-tolerant Gmail ingestion pipeline to batch-process 200+ emails/day, enrich alert data via API, and export to CSV; improved throughput 5x using concurrency, caching, and backoff logic
- Developed a high-concurrency API testing tool (Python + aiohttp) with session auth, dynamic CLI filtering, pagination, and Docker CI/CD integration for benchmarking Eagle Eye's core APIs
- Benchmarked core APIs (/alerts, /alertActions, /alertConditionRules, /notifications), automating P95/P99 latency, throughput, and error metrics to establish baselines and surface filter-driven bottlenecks
- Authored 1,000+ unit tests (pytest-django) and 300+ Postman API tests, achieving 100% coverage across core services and validating authentication, role-based permissions, and complex end-to-end workflows
- Conducted API validation audit across 6+ endpoint groups achieving 75% compliance rate; identified critical PATCH validation gap in alertConditionRules endpoint allowing removal of required fields in production
- Validated platform migration affecting 10,000+ users; developed automated comparison scripts resolving 15% performance discrepancy and confirming 99.9%+ delivery reliability

### Software Engineer, Yunex Traffic

Austin TX | July 2023 – October 2024

- Developed real-time traffic management software in Python, integrating complex scheduling algorithms and database-backed state handling, improving urban traffic flow efficiency 15% across multiple city networks
- Integrated AI/ML models with TensorFlow and Python for predictive traffic analysis, applying time-series forecasting and anomaly detection to cut peak-hour congestion 20% and reduce operational costs 30%
- Migrated legacy traffic management systems to AWS, redesigning services for cloud-native deployment and leveraging Docker + Kubernetes clusters to modernize infrastructure, improve scalability, and system reliability
- Built a load testing framework simulating 3,000+ intersections, incorporating concurrency modeling and data-driven traffic scenarios to validate reliability and scalability of large-scale traffic management systems

- Designed automated test suite validating real-time traffic signal coordination across 50+ intersections, achieving 95%+ timing accuracy under variable load conditions
- Collaborated with embedded systems team to validate NTCIP protocol compliance across 200+ traffic controllers, reducing field deployment failures by 30%

### Software Engineer, Givelify

Austin TX | May 2020 – July 2023

- Developed and deployed a recommendation engine using Python and PyTorch, elevating user retention by 25% and engagement by 20% through cross-functional collaboration with engineering, product, and quality teams
- Refined search capabilities using Python and React, benefiting 10k+ daily users by boosting search efficiency by 15%
- Implemented geocoding services using Google API and LocationIQ, improving location-based search accuracy for 5,000+ daily users and reducing failed lookups by 40%
- Improved donation workflows by building new API endpoints and streamlining payment integrations, reducing transaction latency by 18% and enhancing reliability of high-volume transactions for thousands of daily users
- Built CI/CD pipelines with Jenkins and Docker, cutting deployment time by 70% and improving release reliability
- Architected and maintained pytest test suite with 85%+ code coverage across donation processing, payment gateway, and user authentication services
- Led migration of monolithic payment service to microservices architecture, improving transaction throughput by 45% and reducing latency by 25%

### Junior Software QA Developer, Nourtek

Dallas, TX | October 2019 – May 2020

- Developed and executed 500+ automated JUnit test cases across 3 major releases, reducing production defects by 20%
- Piloted regression testing using Jira, cementing software stability across 25+ releases
- Built automated regression test framework in Java/JUnit, reducing manual QA cycle from 3 days to 4 hours

### Navy Corpsman, United States Navy

Okinawa, JP | July 2011 – July 2016

- Provided medical care for 4,500+ servicemembers in high-stakes operational environments; supported S-2 Intelligence Division managing security clearances for 3,000+ personnel
- Developed leadership, mission-critical decision-making, and cross-functional coordination skills in a fast-paced military environment

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## EDUCATION

Bachelor of Science in Computer Science

University of Texas at Dallas, Richardson, TX | Aug. 2018 - Dec 2021

Associate of Science in Computer Science

Austin Community College, Austin, TX | Aug. 2016 - Aug 2018